The word response is not constant: variability, task drivers and memory

Research proposal (draft)

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November 4, 2026

Summary

Free-recall studies compare words that are later recalled with words that are forgotten (the subsequent memory effect). The shared response to the word itself is subtracted out and treated as constant. In 362 neurosurgical patients performing delayed free recall, we find that this response varies widely across subjects, across electrodes within a region, and from trial to trial. Part of the trial-to-trial variation follows task variables that also predict recall: high-gamma power falls and alpha power rises across serial positions in every lobe. Removing serial position, list number and the blank interval from the high-gamma response halves the frontal subsequent memory effect, removes it in temporal cortex and the hippocampus, and reduces it by a third in occipital cortex. We propose to characterize where the variability sits, which task variables drive it, and how much of the subsequent memory effect they account for.

Background and gap

Serial position. High-frequency activity during encoding falls across serial positions regardless of later recall, and serial position modulates the subsequent memory effect (iEEG, encoding; Serruya et al., 2012); posterior gamma predicts recall mainly for early items (scalp EEG, encoding; Sederberg et al., 2006). Scalp EEG classifiers match recalled and forgotten words across serial positions for this reason (encoding and retrieval; Li et al., 2024), and in fMRI no signal reliably predicts later memory once serial position, item and other confounds are removed (encoding; Halpern et al., 2023). The large iEEG maps of the subsequent memory effect (encoding; Burke et al., 2014; Long et al., 2014; Solomon et al., 2017) did not adjust for these variables jointly.

List number. List-level high-frequency activity predicts how well a whole list is recalled (iEEG, encoding; Rubinstein et al., 2023), and practice or fatigue explain little of list-to-list recall variability (behaviour; Kahana et al., 2018). We found no iEEG study of how list number changes the word response itself.

Blank interval. Longer gaps before an item improve its recall (behaviour; Brown et al., 2006). Responses after a longer wait can be larger because attention is directed to the expected moment (scalp EEG, cued timing; Correa et al., 2006) or because the response has recovered more from the previous stimulus; expected stimuli can also give smaller responses (MEG, auditory; Todorovic

and de Lange, 2012). Whether the blank interval changes the iEEG word response, and in which direction, is untested.

Heterogeneity. High-frequency activity is suppressed in default-mode regions during tasks (iEEG, visual search vs. rest; Ossandón et al., 2011), and activation order is consistent across subjects while individual sites increase or decrease (iEEG, arithmetic vs. rest; Pinheiro-Chagas et al., 2024). Existing large iEEG atlases map recalled vs. forgotten words; none characterizes the word-on vs. word-off response, its electrode-level heterogeneity, or its trial-level drivers.

Data and existing results

FR1, catFR1 and pyFR (362 subjects, 1009 sessions). Each word (word on, 0–600 ms) is compared with the blank screen before it (word off, −700 to −100 ms). Bands: alpha (8–13 Hz, multitaper) and high gamma (70–150 Hz, Hilbert envelopes in sub-bands). The word-onset paper establishes the average response: occipital alpha decreases and high-gamma increases, a weaker version in limbic regions and the hippocampus, and a frontal alpha increase (Fig. 1); high-gamma latencies from ∼120 ms in lateral occipital cortex to ∼320 ms in orbitofrontal cortex; a larger occipital response after longer waits (Fig. 5); weak recall prediction in every lobe (AUC $\leq$ 0.53; Fig. 7); and power changes that are reliable within subjects across sessions (median $r = 0.67$ alpha, 0.81 high gamma; Fig. 8).

Aim 1. Where does the variability sit?

Preliminary. The subject distributions are wide in every region (Fig. 2). Splitting the electrode-level Cohen's d into between-subject and within-subject (between-electrode) parts, subjects account for only 12–37% of the variance in high gamma (frontal 13%, temporal 12%, occipital 25%, hippocampus 37%) and 25–36% in alpha; most of it lies between electrodes of the same region within a subject. Regions also mix directions (Fig. 3): 40% of occipital electrodes increase in high gamma and 3% decrease, while in frontal cortex 5% increase and 4% decrease. In alpha, 21% of occipital electrodes decrease but 11% increase.

Plan. (1) Decompose the response into subject, electrode, session and trial components with a variance-components model, per region and time window. (2) Locate increasing and decreasing electrodes in MNI space within regions (e.g. frontal eye fields and lateral prefrontal cortex vs. the rest of frontal cortex) and test whether they cluster anatomically or by recording type (depth vs. surface). (3) Test whether a subject's alpha and high-gamma responses covary across electrodes (do electrodes with larger alpha decreases show larger high-gamma increases?), which separates real heterogeneity from sampling noise.

Aim 2. Which task variables drive trial-to-trial changes?

Preliminary. Per subject and lobe, we regressed each trial's word-on power (log, z-scored within session, outlier trials removed) on serial position, list number and blank interval (Fig. 4). High gamma falls with serial position in every lobe (occipital −0.042 SD per SD of serial position, temporal −0.032, frontal −0.020; all $q < .05$, 254–339 subjects) and alpha rises (0.009–0.021). Alpha also rises across lists in most lobes, consistent with habituation over a session. The blank interval

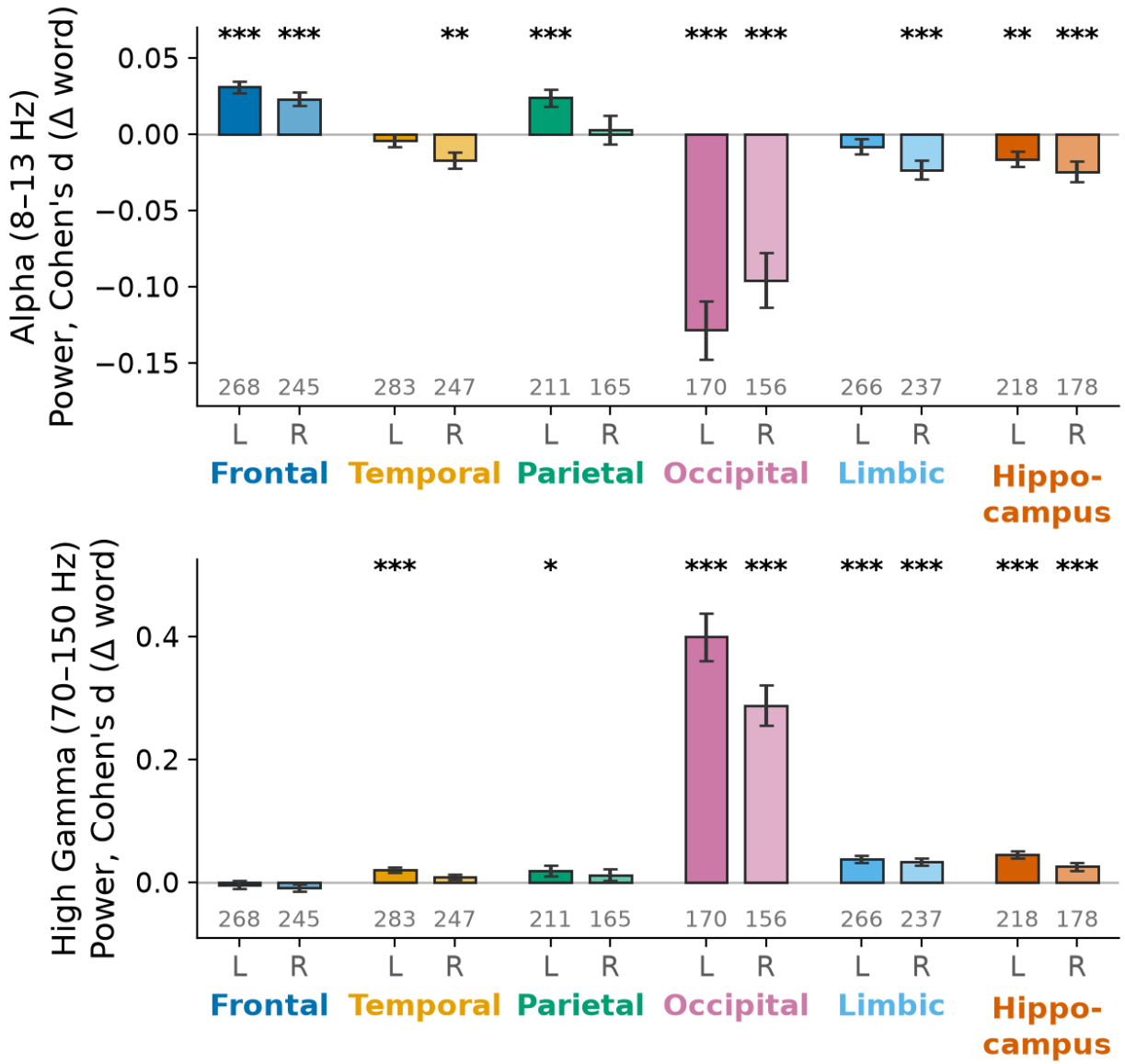


Figure 1: Average word-onset power change per region (word on vs. word off; mean ± SEM across subjects).

had small effects: occipital high gamma +0.007 and frontal alpha +0.006. Together the three variables explain 1–2% of trial-to-trial variance (median R^2), so most trial variation is unexplained by them. Serial position 1 is not in the data: the first word follows the countdown and has no blank-screen baseline.

Plan. (1) Resolve the effects in time and test them on latency as well as magnitude. (2) Separate temporal expectation from recovery after the previous word: add the previous word's response size and its offset time as covariates, and compare a hazard-rate model (expectation grows with waiting) with an adaptation model (recovery with time since the previous word). (3) Test whether serial-position effects shrink over lists and sessions (familiarity with the task). (4) Model the unexplained trial variance: pre-stimulus alpha and high gamma as predictors of the response (state before the word).

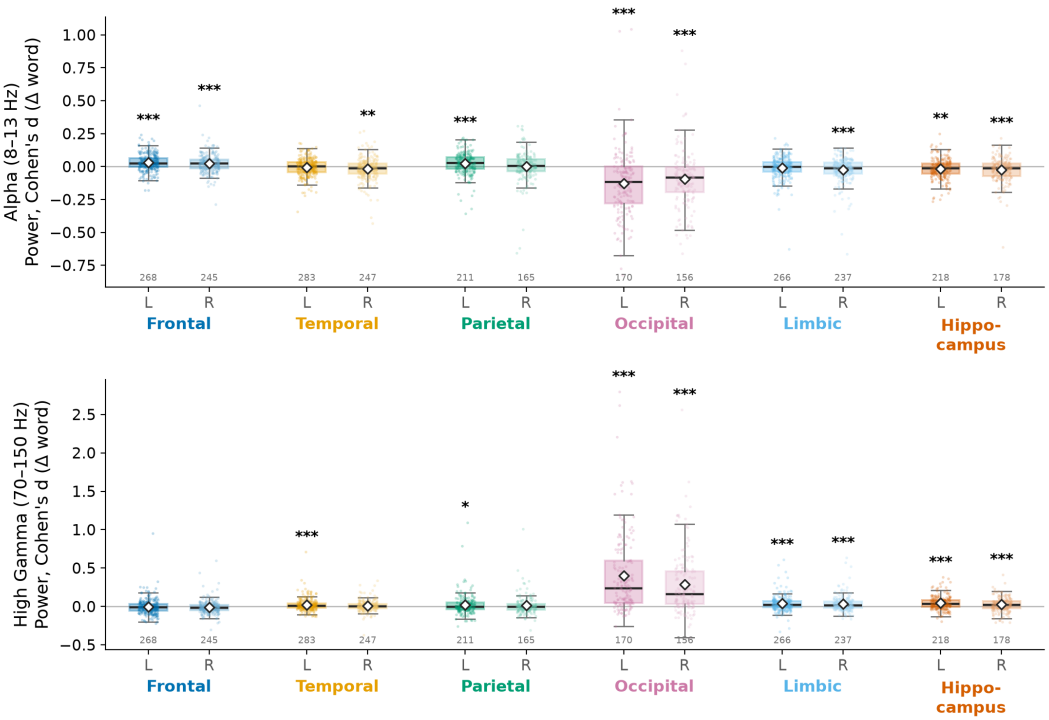


Figure 2: Word-onset power change per region, distributions across subjects.

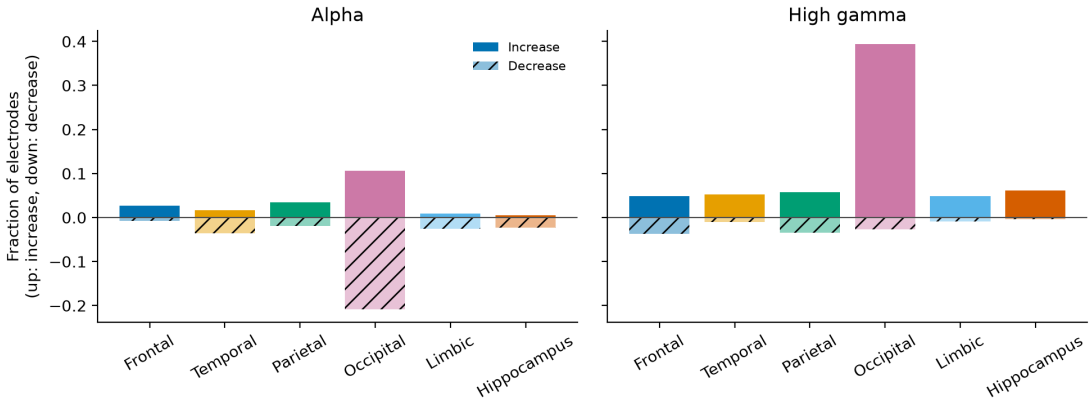


Figure 3: Fraction of electrodes whose power significantly increases (up) or decreases (down) at word onset (per session: Welch t on log power, word on vs. word off, BH-FDR across the session's electrodes).

Aim 3. How much of the subsequent memory effect do task variables explain?

Preliminary. Serial position and list number predict recall across subjects (logistic $\beta = -0.19$ and -0.09 per SD, 361 subjects, both $p < 10^{-6}$); the blank interval does not ($p = .47$; the interval varies only by a few hundred ms). Both also change the word response (Aim 2), so they can create a subsequent memory effect without any memory-specific signal. Removing them from each trial's response (Fig. 6) reduced the high-gamma subsequent memory effect in frontal cortex from $d = 0.030$ to 0.014, in temporal cortex from 0.020 to 0.001, in the hippocampus from 0.014 to 0.001,

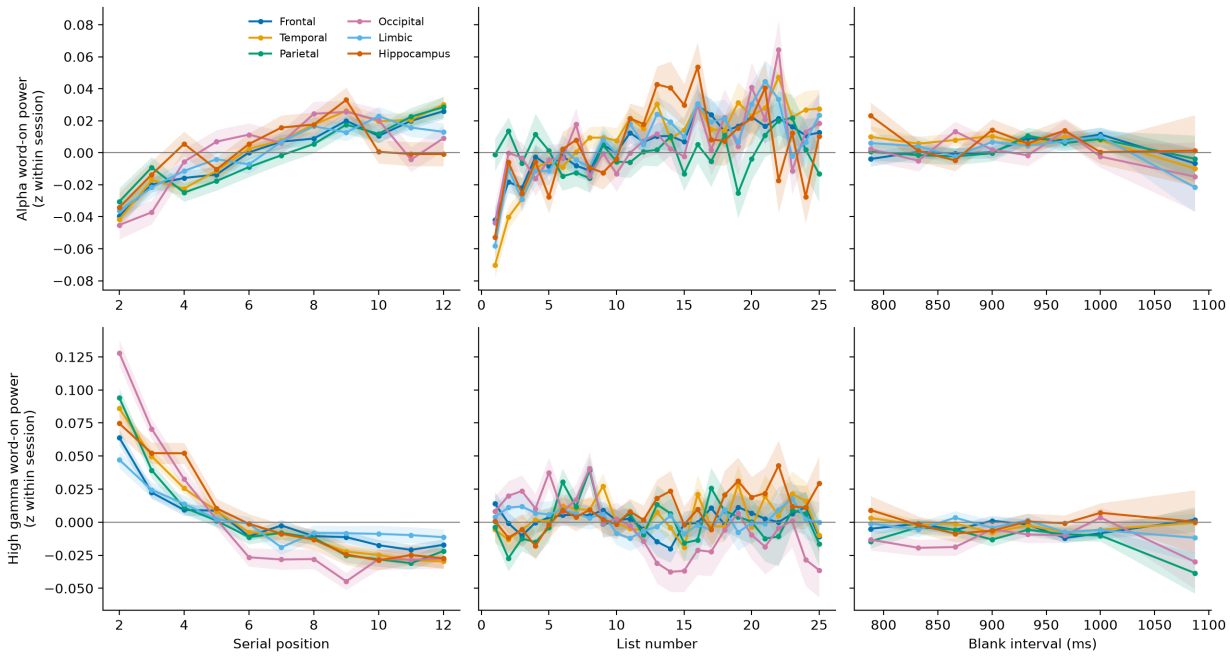


Figure 4: Word-on power (z within session) by serial position, list number and blank interval (quantile bins); mean ± SEM across subjects per lobe. Serial positions 2–12; scored lists only.

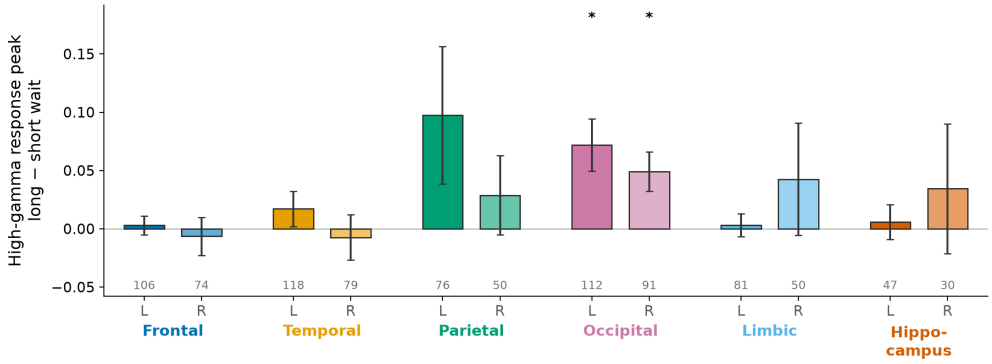


Figure 5: Occipital high-gamma response after long vs. short waits (from the word-onset paper).

and in occipital cortex from 0.098 to 0.067 (paired t, all $p < .01$). The occipital alpha effect shrank from -0.051 to -0.037. The occipital effect survives, while the frontal and temporal effects are largely accounted for by where the word fell in the list.

Plan. (1) Repeat with the time-resolved classifier (Fig. 7) with covariates removed, and with recalled and forgotten words matched on serial position and list (as Li et al., 2024). (2) Test whether the blank interval affects recall of the next or the current word, and whether the word response mediates it. (3) Ask whether electrodes that are suppressed at word onset (Aim 1) carry a different subsequent memory effect from those that are activated, in frontal cortex in particular.

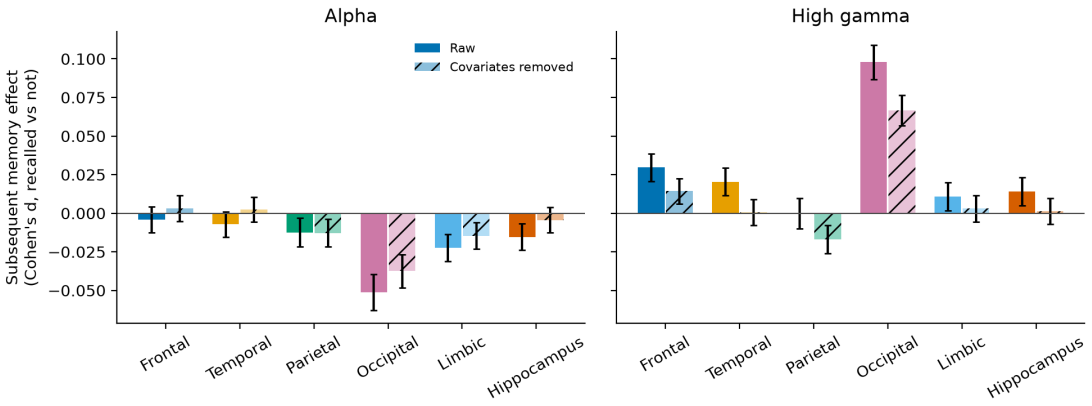


Figure 6: Subsequent memory effect (Cohen's d, recalled vs. not recalled, word on 0–600 ms) per lobe, raw and after regressing each trial's response on serial position (linear and positions 2–3), list number and blank interval within subject; mean ± SEM across subjects.

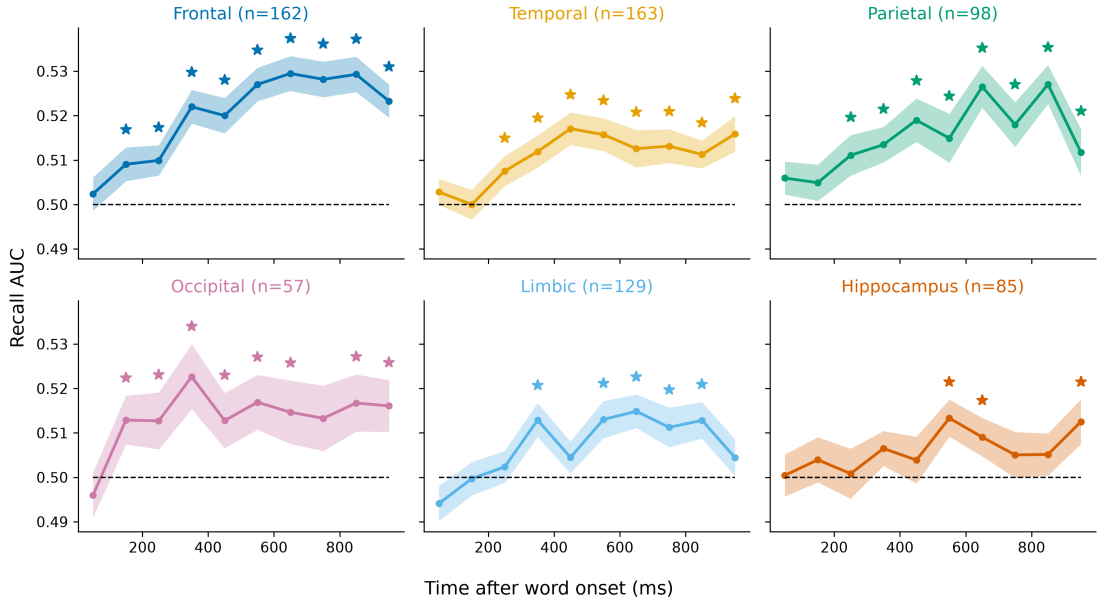


Figure 7: Recall prediction from the high-gamma response (encoding contrast; per-session AUC, mean ± SEM across subjects; from the word-onset paper).

Reliability

Power changes are reliable within subjects across sessions, while synchrony changes are not (Fig. 8). The proposal therefore focuses on power; connectivity enters only where it is reliable (the occipital network).

Open issues

- **Artifact trials.** The current pipeline has no trial-level artifact rejection. One trial in one subject (a broadband transient on 82 of 148 electrodes) produces a spike at 650 ms in

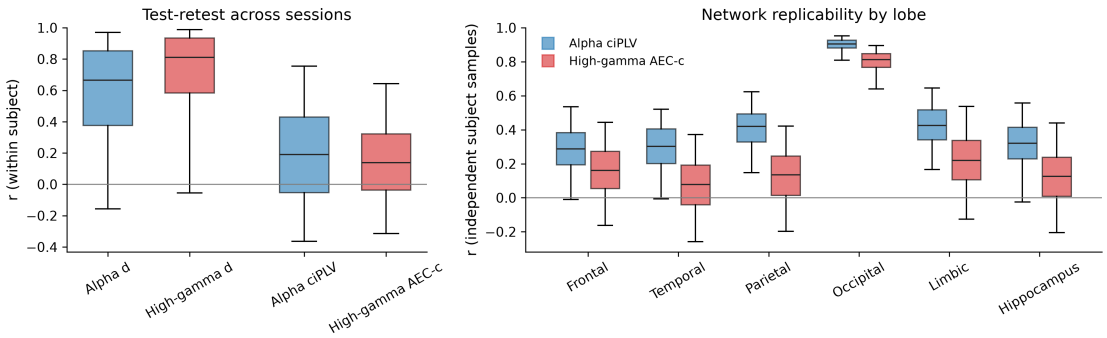


Figure 8: Session-to-session reliability within subjects, and replicability across subjects (from the word-onset paper).

the supramarginal gyrus group average. The preliminary results drop robust outliers per electrode; the main pipeline needs the same.

- **Serial position 1** is excluded by the baseline design; primacy is measured from position 2.
- **Blank intervals** span a narrow range, which limits the expectation analysis; RepFR1 or ltpFR2, with different timing, would extend it.
- **Extensions:** eye tracking during word presentation (Kucewicz datasets) and repetition suppression (RepFR1).

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